

Water Level Indicator

❖ Summery

In this project we will be building a water level indicator using BC547 Transistor with three levels that consist of inside a glass with the help of 3 LED and a buzzer to indicate that the glass is full.

❖ Objectives

- I. To learn the working of the water level indicator.
- II. Measure the water level when the circuits indicate when the glass its half and full.
- III. To learn how to build simple circuits.

❖ Industry-Based Applications

The water level indicator circuits are used in factories, chemical plants, and electrical substations and in other liquid storage systems. There are many possible uses for this simple system, examples include monitoring a sump pit (to control pump activation), rainfall detection, and leakage detection. Electronic water level circuits have the capability of alerting if there is a water leak somewhere in the factory. When the water level is too high or too low or exceeds the higher limit, it can detect the water level easily by hearing an alarm sound or from different colours of a light bulb. We can also measure the fuel level in motor vehicles and the liquid level containers which are huge in the companies.

❖ Project Methodology

The circuit is designed to indicate three levels of water stored in the glass: low but not empty, half and full but not overflowing. When there is no water in the glass, all the LEDs are off as an indication that

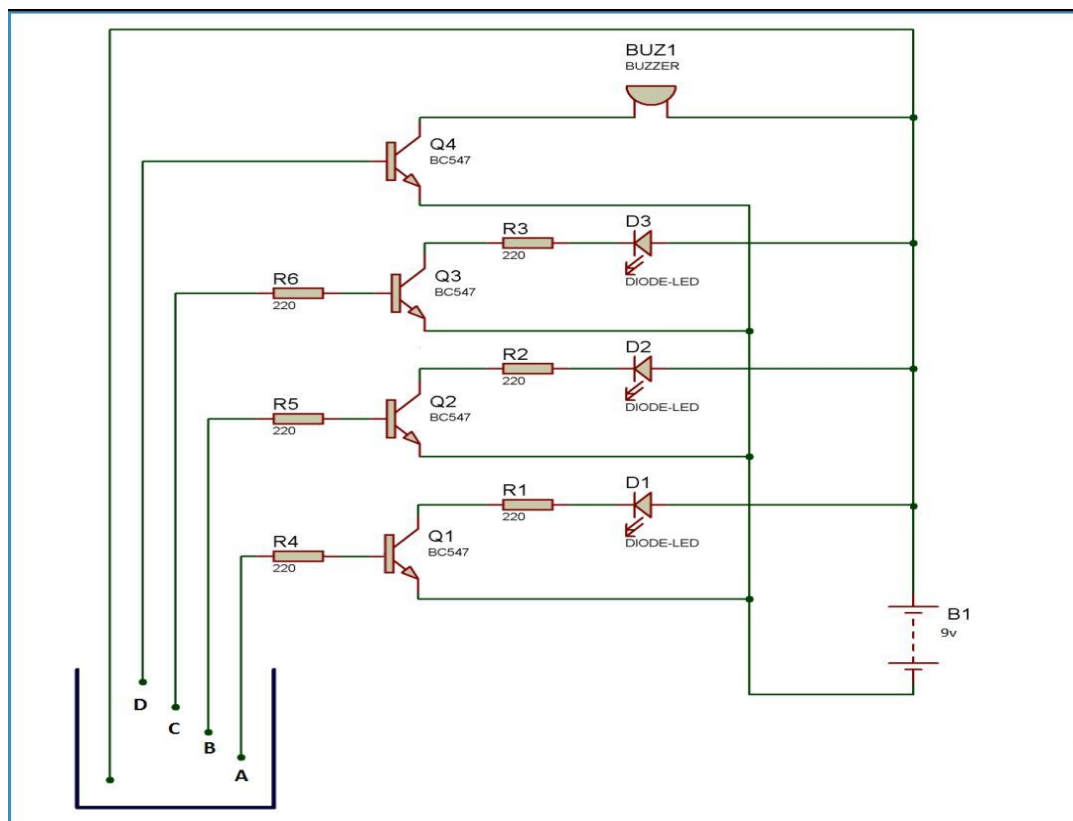
the glass is completely empty. When water level increases and touches the sensor, the first LED will glow indicating that there is water within the glass. As the water level continues to rise and reaches half the glass, second LED will glow. When the water in the glass rises to full an alarm is made by the buzzer as an indication that the glass is full.

❖ Project Procedures:

Components

1. BC547 Transistors (x3)
2. 330 ohms resistors (x3)
3. LEDs (x3)
4. Buzzer
5. Some connecting wires
6. 9V Battery with Battery Cap
7. Breadboard

❖ Circuit Diagram



❖ **Constructions of the circuit**

1. In the diagram Q1, Q2.. are transistors with an arrow showing emitter.
2. All the emitters are connected to +ve terminal of the battery and all the collectors are connected to LED (**except 3rd which is connected to the buzzer**) via **330 Ω** resistor.
3. The base of all the transistors is dipped into the water via a **330 Ω** resistor.
4. The 3rd transistor is used to power the buzzer when the water level rises to its sensor.

❖ **Working of water level indicator**

When we power the circuit via a 9V or 5V battery the circuit is in off state there is no current flow through the transistors and the remaining circuit.

When we dip the base terminals of all the transistors one by one in water and also add the +Ve terminal of the battery to water glass, a current in micro Amperes starts flowing via base terminal. This leads to the switching of the transistors to ON state. As the water level in the mug or tank rises the other base terminals also get connected to the battery via water and thus other LEDs and buzzer finally start glowing.

This is how the circuit works. We can use it at our house.

❖ **Advantages**

1. Easy to installation
2. Minimal maintenance
3. Compact design
4. Automatically adjust the water level

5. Sends an alert to let you know water is too high or too low.

❖ Disadvantages

1. No warranty and guarantee
2. Water level controls need to be replaced every 3 years

❖ Applications

1. It is used in water tanks to controls water level
2. Water level control
3. Tsunami warnings and sea level monitoring etc.

❖ Conclusion

In this water level project, I learned water level control system. How to control water using electronic circuit. So, I really enjoy to making this project.

Thank you!